

Optimal Sizing of Solar, Wind, and Battery Storage to Compensate for Nuclear Decommissioning in France

Matthieu Hautsch, Anna-Sherazade Sylla

1. Introduction

Nuclear power is the backbone of France’s electricity system. In 2024, 67% of French electricity came from nuclear plants, with 57 reactors operating across 18 sites and a total installed nuclear capacity of 61.4 GW — the highest nuclear share of any country in the world [1, 2]. This unique energy profile has allowed France to maintain one of the lowest carbon intensities in Europe.

However, this fleet is aging. The majority of France’s reactors were connected to the grid between 1977 and 1993, and were originally designed for a 40-year operational lifetime. The average age of the current fleet is 39 years, and the oldest still-operating reactors date back to 1979. In 2021, the *Autorité de Sûreté Nucléaire* officially extended the operating license of the oldest reactors from 40 to 50 years [3]. Further extensions to 60 years are currently under consideration [4].

If a strict 50-year lifetime were applied with no new nuclear construction, the consequences would be substantial: based on IAEA PRIS data [5], 9.92 GW of nuclear capacity would need to be retired by 2030. Meanwhile, France’s combined share of wind and solar generation stands at only 13%, well below the EU average of 30%, leaving a significant gap to fill [1].

This raises a central capacity expansion problem: what mix of solar, wind, and storage minimizes system costs to compensate for the projected nuclear capacity loss?

2. Methods

2.1 Main Assumptions

System boundary and temporal representation. We model France as an isolated system, no cross-border imports or exports are permitted at any hour, and partition the metropolitan territory into twelve administrative regions that correspond to the spatial resolution of the RTE generation-mix and consumption data [6, 7].

Solving a full year at hourly resolution (8,760 time steps) with integer transmission variables would be computationally intensive. We therefore compress the year into 10 representative days obtained by k -medoids clustering of the 2024 hourly profiles, weighting equally on load shape and generation mix. Each of the 366 calendar days is assigned to its closest representative day, so that the full year remains covered. The ten resulting clusters and their consumption envelopes are shown in Figure 1.

Demand and residual generation. Hourly regional generation and consumption profiles for the year 2024 are sourced from the RTE eCO2mix regional dataset [7], which provides, for each of the twelve regions and each hour of the year, the observed output broken down by technology as well as the regional load. France is a net exporter of electricity, so total generation typically exceeds domestic consumption, as visible in Figure 2. Demand profiles are scaled upward by +8 % to reflect the future demand growth expected by 2030 under RTE electrification scenarios [8].

To determine which nuclear generation must be replaced, we identify all reactors scheduled for retirement under a 50-year lifetime assumption using the IAEA Power Reactor Information System [5]. For each representative day, the hourly output of the retired reactors is extracted directly from the RTE API, and subtracted from the total observed regional generation profile. This yields a residual generation time series $\tilde{G}_{r,d,t}$ — what the remaining non-retired fleet can be expected to produce in each region r , representative day d , and hour t . The new solar, wind,

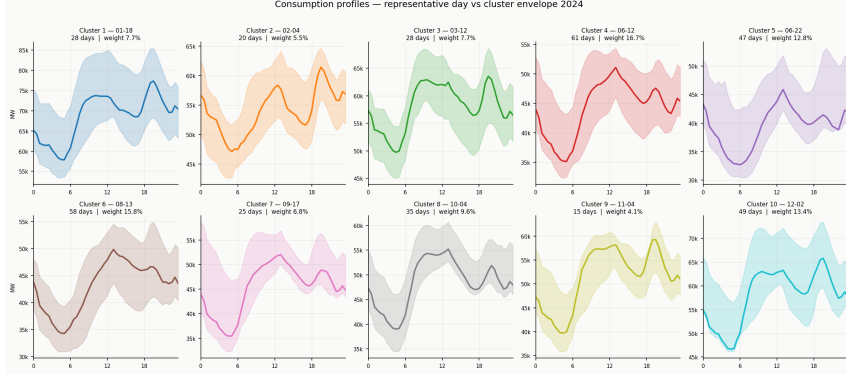


Figure 1: Consumption profiles for each of the ten representative days obtained by k -medoids clustering of the 2024 hourly national load. The solid line shows the medoid (representative day) and the shaded band the cluster envelope.

and battery capacity that the model optimizes must, in aggregate with $\tilde{G}_{r,d,t}$, cover the scaled demand $D_{r,d,t}$ at every hour.

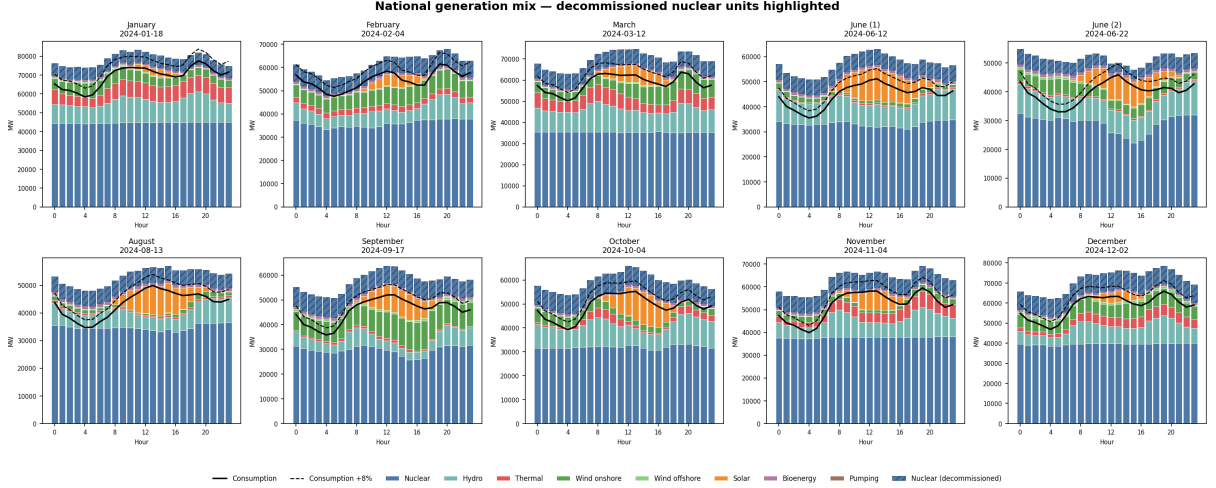


Figure 2: National generation mix for each of the ten representative days.

Renewable capacity factors. The optimal solar PV and onshore wind capacities to install in each region are determined by the model based on hourly, region-level capacity factors. These are sourced from Renewables.ninja [9, 10], which reanalyses ERA5 meteorological data with representative plant configurations, and extracted for each of the ten representative days.

Battery modeling. Batteries are modeled as 4-hour lithium-ion systems; round-trip efficiency losses are neglected for simplicity. To capture multi-week storage dynamics within the representative-day framework, we adopt the two-level state-of-charge formulation of Kotzur et al. [11]. This method imposes an inter-period state of charge on top of the intra-period dynamics, linking the representative days through the sequence of calendar days they represent. This allows the model to capture seasonal storage behavior at a fraction of the computational cost of a full 8,760-hour formulation.

Transmission network. The RTE/ODRE dataset [12] provides line-level data for every aerial and underground transmission circuit in metropolitan France, recorded at the commune level. We

aggregate these lines by region pair to construct a twelve-node inter-regional network, summing the capacity of all circuits connecting each pair of adjacent regions. Individual circuit capacities are assigned following the PyPSA-Eur line-type library [13]: 400 kV overhead corridors are rated at 1 500 MW and underground corridors at 1 000 MW, while 225 kV overhead corridors carry 400 MW and underground corridors 300 MW. Transmission losses are neglected for simplicity. The resulting inter-regional network is shown in Figure 3. Inter-regional distances are computed as distances between region centroids, then multiplied by a tortuosity factor of 1.25 to account for the non-straight routing of real transmission lines [14, 15].

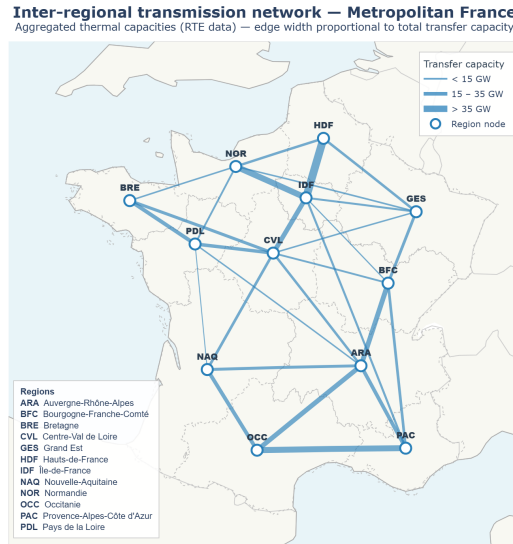


Figure 3: Inter-regional transmission network for metropolitan France.

For new construction, only overhead lines are considered. The RTE *Schéma Décennal de Développement du Réseau* confirms that underground 400 kV AC cables are not a viable option in France as they would require HVDC conversion, and that underground 225 kV cables cost on average 65% more than equivalent overhead lines [17].

Costs. All technologies are costed on an annualized basis, combining capital expenditure (CAPEX) and fixed operations and maintenance (O&M) at a uniform 6% discount rate. For transmission, investment costs are taken from the Danish Energy Agency catalogue for 400 kV lines [16] and derived from the RTE *Schéma Décennal de Développement du Réseau* for 225 kV lines [17]. We used a fixed O&M rate of 1.5%/year of CAPEX [16]. For solar PV, onshore wind, and battery storage, annualized costs are derived from RTE *Futurs énergétiques 2050* [8]. Table 1 summarizes the resulting per-unit annualized figures for all technologies considered.

Table 1: Annualised technology costs used in the optimisation.

	Solar PV	Onshore Wind	Battery (4 h)	225 kV line	400 kV line
Lifetime (years)	25	25	20	40	40
Annualized cost	84.21/kW€	70.87/kW€	25.53€/kWh	101.83€/MW·km	61.10€/MW·km

A summary of all data sources used in this study is provided in Table 2 in the Appendix.

2.2 Mathematical Formulation

The full problem is formulated as a Mixed-Integer Linear Programme (MILP) combining a network flow model governing power exchange between the twelve regional nodes with a time series

model tracking generation, storage, and demand across all representative days at hourly resolution.

2.2.1 Sets and Indices

- $r \in \mathcal{R} = \{1, \dots, 12\}$: regions.
- $d \in \mathcal{D} = \{1, \dots, 10\}$: representative days.
- $t \in \mathcal{T} = \{1, \dots, 24\}$: hours within a day.
- $i \in \mathcal{I} = \{1, \dots, 366\}$: calendar days.
- $(r, r') \in \mathcal{P} \subseteq \mathcal{R}^2$: adjacent region pairs.

2.2.2 Decision Variables

Investment.

- $x_r^{\text{sol}}, x_r^{\text{wnd}} \geq 0$: new generation capacity [MW] of solar PV and onshore wind in region r .
- $x_r^{\text{bat}} \geq 0$: new battery energy capacity [MWh] in region r .
- $y_{r,r'}^{400}, y_{r,r'}^{225} \in \mathbb{Z}_{\geq 0}$: number of new 400 kV and 225 kV circuits added on corridor (r, r') .

Operation.

- $f_{r,r',d,t} \in \mathbb{R}$: net power flow [MW] from r to r' (negative = reverse direction).
- $b_{r,d,t}^+, b_{r,d,t}^- \geq 0$: battery charge and discharge power [MW].
- $e_{r,d,t}^{\text{intra}}$: intra-period state of charge [MWh].
- $e_{r,d}^{-\text{intra}}, e_{r,d}^{+\text{intra}}$: minimum and maximum intra-period SOC over the representative day.
- $e_{r,i}^{\text{inter}} \geq 0$: inter-period (calendar-day level) state of charge [MWh].

2.2.3 Objective Function

The model minimises the total annualised system cost, comprising generation, storage, and transmission investment:

$$\min \sum_r \left[c^{\text{sol}} x_r^{\text{sol}} + c^{\text{wnd}} x_r^{\text{wnd}} + c^{\text{bat}} x_r^{\text{bat}} \right] + \sum_{(r,r') \in \mathcal{P}} \left[y_{r,r'}^{400} c^{400} d_{r,r'} F^{400} + y_{r,r'}^{225} c^{225} d_{r,r'} F^{225} \right], \quad (1)$$

where $d_{r,r'}$ is the inter-regional distance multiplied by the tortuosity factor (1.25), and F^{400}, F^{225} are the MW ratings of a single new 400 kV and 225 kV circuit, respectively.

2.2.4 Constraints

Energy balance. For every region r , representative day d , and hour t , total supply must meet or exceed demand:

$$\tilde{G}_{r,d,t} + \text{CF}_{r,d,t}^{\text{sol}} x_r^{\text{sol}} + \text{CF}_{r,d,t}^{\text{wnd}} x_r^{\text{wnd}} + b_{r,d,t}^- - b_{r,d,t}^+ + \sum_{r':(r',r) \in \mathcal{P}} f_{r',r,d,t} - \sum_{r':(r,r') \in \mathcal{P}} f_{r,r',d,t} \geq D_{r,d,t}, \quad (2)$$

where $\tilde{G}_{r,d,t}$ is the residual generation after nuclear retirement, $\text{CF}_{r,d,t}^{\text{sol}}$ and $\text{CF}_{r,d,t}^{\text{wnd}}$ are the hourly capacity factors, and $D_{r,d,t}$ is the demand.

Battery dynamics - Kotzur et al. 2018 [11]. The state of charge is decomposed into two layers. $e_{r,d,t}^{\text{intra}}$ is the intra-period SOC accumulated within representative day d , initialized at zero at the start of each day; it captures the within-day charging and discharging profile. $e_{r,i}^{\text{inter}}$ is the inter-period SOC at the start of calendar day i ; it accumulates the net daily SOC changes across the 366 days of the year, enabling energy to be carried from one period to the next. Intra-period dynamics evolve hour by hour as

$$e_{r,d,t}^{\text{intra}} = e_{r,d,t-1}^{\text{intra}} + b_{r,d,t}^+ - b_{r,d,t}^-, \quad e_{r,d,0}^{\text{intra}} = 0, \quad (3)$$

$$e_{r,d}^{-\text{intra}} \leq e_{r,d,t}^{\text{intra}} \leq e_{r,d}^{+\text{intra}}, \quad b_{r,d,t}^{\pm} \leq \frac{x_r^{\text{bat}}}{4}. \quad (4)$$

The net transfer at the end of each representative day is then accumulated across calendar days,

$$\sum_{d \in \mathcal{D}} n_d e_{r,d,T}^{\text{intra}} = 0, \quad (5)$$

$$e_{r,i+1}^{\text{inter}} = e_{r,i}^{\text{inter}} + e_{r,d(i),T}^{\text{intra}}, \quad (6)$$

$$0 \leq e_{r,i}^{\text{inter}} + e_{r,d(i)}^{+\text{intra}} \leq x_r^{\text{bat}}, \quad (7)$$

$$0 \leq e_{r,i}^{\text{inter}} + e_{r,d(i)}^{-\text{intra}}. \quad (8)$$

where n_d is the number of calendar days assigned to representative day d , and $d(i)$ maps calendar day i to its representative day.

Transmission capacity. Power flows on each corridor are bounded by the sum of pre-existing and newly built capacity:

$$|f_{r,r',d,t}| \leq \bar{T}_{r,r'} + y_{r,r'}^{400} F^{400} + y_{r,r'}^{225} F^{225}, \quad (9)$$

where $\bar{T}_{r,r'}$ is the existing transfer capacity on corridor (r, r') .

3. Results

The optimization model is solved using the MILP solver HiGHS [20]. Since the problem is formulated as a mixed-integer linear program, all reported solutions are globally optimal up to the specified optimality tolerance.

3.1 Baseline scenario

We first consider a baseline scenario with an electricity demand increase of 8% by 2030 and the retirement of approximately 10 GW of nuclear capacity. The existing transmission network is represented explicitly, and investments in additional transmission capacity are allowed.

The optimal solution yields a total annualized system cost of 2.98 B€/yr. To replace the retired nuclear generation and satisfy the increased demand, the model selects 12.6 GW of new wind capacity, primarily located in the Hauts-de-France region, with additional installations in Provence-Alpes-Côte d’Azur (PACA). No utility-scale solar PV capacity is selected, reflecting the combination of lower capacity factors and less favorable economics relative to wind under the baseline assumptions.

The model also deploys 32.2 GWh (8 GW of discharge power) of four-hour battery storage, concentrated in Hauts-de-France. This storage capacity complements the large wind installations

by shifting excess generation to periods of lower wind output and helping maintain system reliability.

Interestingly, the model does not invest in any new transmission infrastructure. This result suggests that the existing French transmission network is sufficiently robust to accommodate both the projected demand growth and the replacement of the retired nuclear capacity without requiring major grid expansions.

The geographical distribution of new investments is shown in Figure 4, together with the locations of the decommissioned nuclear reactors.

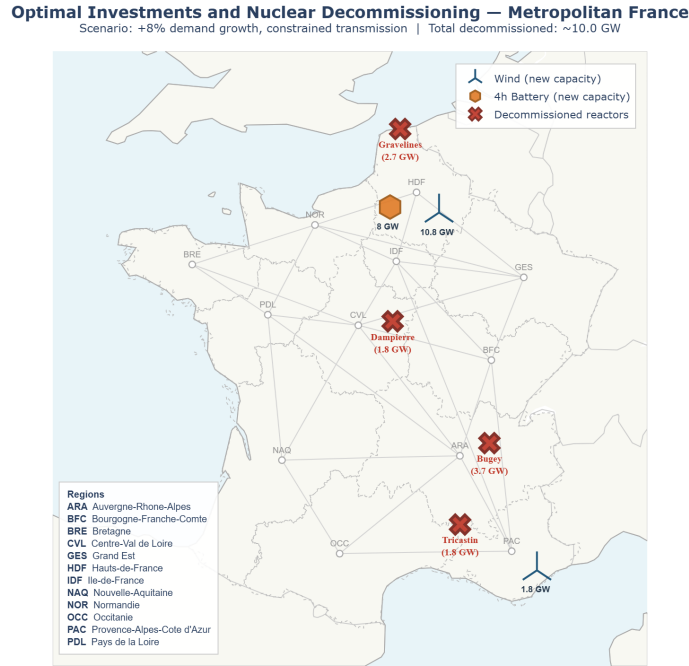


Figure 4: Locations of the new capacity built.

3.2 Comparison of scenarios and fossil-free pathways

The baseline model relies on generation data from 2024. As a result, the Flamanville 3 EPR reactor is not represented in the original dataset, since it only entered commercial operation in December 2024. To assess its contribution to the French electricity system by 2030, we introduce an additional scenario in which Flamanville 3 is included as a fixed nuclear generation source in the Normandie region.

The reactor is modeled with a nominal capacity of 1.63 GW and an assumed capacity factor of 75%, corresponding to an annual production of approximately 10.7 TWh. This value is higher than the current French fleet average but remains conservative for a new EPR unit expected to operate under normal operating conditions by 2030 [8, 21, 22]. Because nuclear generation is treated as an exogenous parameter rather than a dispatchable decision variable, its output is assumed to be constant throughout the year, a common simplification in long-term capacity expansion studies.

We also investigate a fossil-free scenario in which the remaining gas-fired generation in France is removed and replaced by renewable generation and storage resources. Although gas currently accounts for a relatively small share of annual electricity production, it provides valuable flexibility during periods of high demand and low renewable output. Combining this assumption with the inclusion of Flamanville 3 yields four scenarios, whose results are summarized in Figure 5.

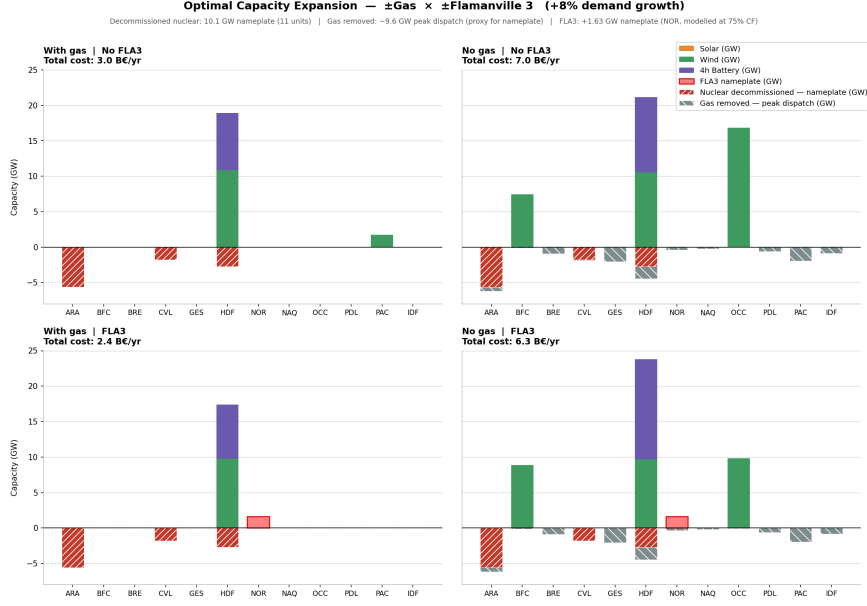


Figure 5: Comparison of optimal system costs and capacity investments across the four modeled scenarios.

Several trends emerge from these results. First, eliminating the remaining fossil generation substantially increases system costs. The optimal system cost rises by a factor of approximately 2.4 (from 2,975 M€/yr to 7,031 M€/yr), while the required wind capacity increases by nearly a factor of 2.8. This result highlights the important role currently played by gas-fired generation in providing operational flexibility, despite its limited contribution to total annual energy production.

Second, the addition of Flamanville 3 reduces investment requirements in both fossil-allowed and fossil-free scenarios. When fossil backup remains available, the reactor decreases total system costs by approximately 525 M€/yr (from 2,975 to 2,450 M€/yr) and reduces the required wind capacity by about 18%. In the fossil-free case, Flamanville 3 reduces the optimal cost from 7,031 to 6,280 M€/yr, a saving of 751 M€/yr. The additional low-carbon generation from the EPR directly offsets energy that would otherwise need to be provided by new renewable capacity.

Interestingly, in the fossil-free case, adding Flamanville 3 increases battery deployment despite reducing total system cost and renewable capacity requirements. This result stems from the flat output profile assumed for nuclear generation. While the additional baseload energy reduces overall renewable capacity requirements, it does not necessarily coincide with periods of peak demand. The resulting temporal mismatches increase the need for storage to shift energy across time and maintain system reliability.

Finally, the geographical distribution of renewable investments changes in the absence of fossil generation. While Hauts-de-France remains the dominant wind-producing region, investments in PACA disappear in fossil-free scenarios and are replaced by larger deployments in Occitanie and Bourgogne-Franche-Comté, reflecting the need to exploit a broader set of renewable resources when the system must operate without gas-fired flexibility.

3.3 Illustrative comparison with nuclear replacement costs

To contextualize the cost of renewable-based system expansion, we compare the annualized system cost of the optimized renewable-based solutions with an illustrative nuclear replacement benchmark based on Flamanville 3 (EPR).

Using the realized investment cost of Flamanville 3 (13.2 B€ for 1.63 GW), replacing 10 GW of nuclear capacity would require approximately six equivalent units. Assuming a 6% discount rate over a 60-year lifetime, the corresponding annualized cost is approximately 4.9 B€/yr.

This value is higher than the baseline optimized system cost of 2.98 B€/yr. This comparison should be interpreted as an order-of-magnitude benchmark, as it does not account for fuel costs, operational flexibility, or system-wide integration effects.

4. Sensitivity Analysis

All sensitivity analyses are solved to global optimality using the HiGHS MILP solver. Variations in the results therefore reflect genuine changes in the optimal system configuration rather than solver heuristics or local optima.

We investigate two sources of uncertainty: future electricity demand growth and the cost of renewable generation and storage technologies.

4.1 Sensitivity to Demand Growth

Demand growth is varied from 0% to 20%, with 8% corresponding to the baseline scenario. Results are presented in Figure 6.

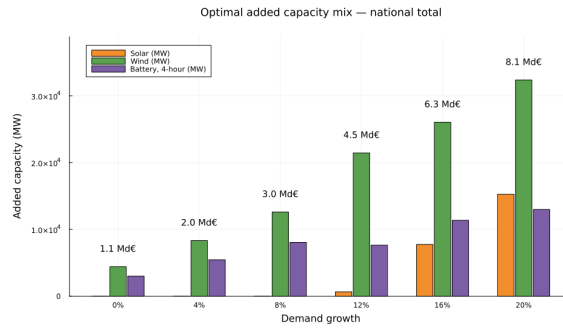


Figure 6: Optimal capacity additions and annualized system cost as a function of demand growth.

System cost increases almost linearly for demand growth rates up to approximately 10%. Over this range, the optimizer relies exclusively on wind power and battery storage. Wind capacity grows from 4.4 GW to 16.5 GW, while battery deployment increases proportionally. No solar PV is selected.

A transition occurs at approximately 12% demand growth, where solar PV enters the optimal solution for the first time. Beyond this threshold, both wind and solar capacity expand rapidly, and total system cost increases more steeply, reaching 8.1 B€/yr at 20% demand growth.

The baseline scenario remains below this transition point. As a result, the conclusion that wind is the preferred generation technology is robust to moderate uncertainty in demand forecasts. However, significantly higher demand growth would require a more diversified renewable portfolio and substantially larger investments.

Notably, no new transmission capacity is built in any of the eleven demand-growth scenarios, indicating that the existing French grid can accommodate substantial electrification without major reinforcement.

4.2 Sensitivity to Infrastructure Cost

To assess how robust the optimal investment mix is to uncertainty in technology costs, we vary wind and battery costs between 70 % and 130 % of their baseline values, generating a 5×5 grid of 25 model runs. Results are presented in Figure 7.

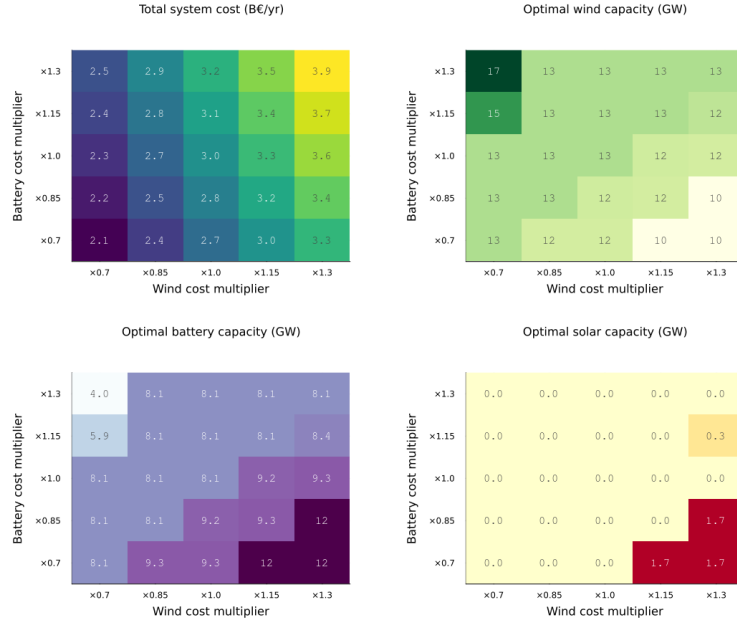


Figure 7: Optimal system cost and capacity deployment for different wind and battery cost assumptions.

Total annualized system cost ranges from 2.08 B€/yr to 3.87 B€/yr. Wind cost is the dominant driver of system economics: varying wind costs produce substantially larger cost changes than varying battery costs, reflecting the central role of wind generation in all optimal solutions.

A clear substitution effect exists between wind and storage. When batteries are inexpensive, the optimizer installs less wind capacity and relies more heavily on storage. Conversely, when wind is relatively cheap, the model favors additional wind generation and reduces battery deployment. Along this trade-off frontier, total system cost remains relatively stable.

Solar PV appears only in a few extreme cases characterized by high wind costs and low battery costs. In all other scenarios, wind remains the preferred renewable technology.

Overall, the baseline result seems robust. Across almost all cost combinations, the optimal solution remains dominated by wind generation supported by battery storage, with little or no solar deployment. As in the demand-growth analysis, no new transmission capacity is built in any of the 25 sensitivity runs.

5. Conclusion

5.1 Discussion

The results highlight several consistent structural features of the French power system under nuclear decommissioning. Across all scenarios and sensitivity analyses, wind power combined with battery storage emerges as the dominant least-cost solution, while solar PV plays only a marginal role under baseline assumptions.

A key result is the absence of new transmission investment in all scenarios. This suggests that,

under the modeled spatial aggregation and cost assumptions, the existing French transmission network is sufficiently strong to accommodate both demand growth and large-scale renewable integration. However, this result should be interpreted cautiously given the aggregated regional representation of the grid.

The analysis also highlights the critical role of gas-fired generation as a flexibility resource. Although its contribution to annual energy is relatively small, its removal in the fossil-free scenario leads to a disproportionate increase in system cost (approximately $2.4\times$) and a near-tripling of required wind capacity. This confirms that dispatchable capacity, even at low utilization, has a strong system value in balancing variability.

Nuclear generation exhibits a dual effect. The inclusion of Flamanville 3 reduces total system cost and renewable capacity requirements, but in the fossil-free case it also increases battery deployment. This result is driven by the assumption of flat nuclear output, which improves energy adequacy but does not contribute to intra-day flexibility.

These results reflect an optimization under strong structural assumptions (notably no cross-border trade, simplified dispatch, and reduced temporal representation), and should therefore be interpreted as indicative capacity expansion signals rather than predictive deployment outcomes.

Overall, under the current modeling assumptions, system decarbonization is primarily achieved through a wind-dominated expansion complemented by storage, with limited structural pressure for transmission reinforcement.

5.2 Main limitations

Several modeling assumptions may affect the quantitative results:

- **Spatial resolution:** The model aggregates France into 12 regions, which may underestimate local congestion and overestimate transmission adequacy.
- **Temporal representation:** The use of 10 representative days may smooth extreme events and underrepresent rare stress periods.
- **Dispatch simplification:** Existing generation is treated as exogenous residual production rather than fully co-optimized dispatch.
- **Nuclear lifetime assumption:** A strict 50-year lifetime is assumed, whereas extensions to 60 years are actively discussed in France [4].
- **No cross-border exchanges:** The model excludes imports/exports, likely overstating domestic capacity requirements.

5.3 Future work

Future extensions of this work could improve realism and policy relevance in several directions:

- Endogenous unit commitment to jointly optimize investment and dispatch decisions.
- Explicit reliability constraints (e.g., reserve margins or loss-of-load probability).
- Higher spatial resolution, including sub-regional or nodal transmission modeling.
- Integration of offshore wind and cross-border European power exchanges.
- Extension to longer-term horizons with phased nuclear retirements and new build scenarios.

References

- [1] EDF, *Le nucléaire en chiffres*, 2024. <https://www.edf.fr/groupe-edf/comprendre/production/nucleaire/nucleaire-en-chiffres>
- [2] IEA, *Energy System of France*, International Energy Agency, 2024. <https://www.iea.org/countries/france>
- [3] France 24, *Officialisation de la prolongation des plus vieux réacteurs nucléaires français de 40 à 50 ans*, 25 février 2021. <https://www.france24.com/fr/info-en-continu/20210225-officialisation-de-la-prolongation-des-plus-vieux-r%C3%A9acteurs-nucl%C3%A9aires-fran%C3%A7ais-de-40-%C3%A0-50-ans>
- [4] IAEA, *Long-Term Operation of Nuclear Power Plants – France*, INIS Collection. <https://inis.iaea.org/records/beeb9-kc067>
- [5] IAEA, *Power Reactor Information System (PRIS) – France*, 2024. <https://pris.iaea.org/pris/CountryStatistics/CountryDetails.aspx?current=FR>
- [6] RTE, *eCO2mix – Consommation et production nationales*, 2024. <https://www.rte-france.com/donnees-publications/eco2mix-donnees-temps-reel/consommation-france>
- [7] RTE / ODRE, *eCO2mix – Consommation et production régionales*, 2024. <https://odre.opendatasoft.com/explore/dataset/eco2mix-regional-cons-def/>
- [8] RTE, *Futurs énergétiques 2050 – Réactualisation 2026*, 2026. <https://www.rte-france.com/donnees-publications/etudes-prospectives/futurs-energetique-2050>
- [9] Pfenninger, S. and Staffell, I., *Renewables.ninja – Solar capacity factors*, 2024. <https://www.renewables.ninja/>
- [10] Pfenninger, S. and Staffell, I., *Renewables.ninja – Wind capacity factors*, 2024. <https://www.renewables.ninja/>
- [11] L. Kotzur, P. Markewitz, M. Robinius, and D. Stolten, *Time series aggregation for energy system design: Modeling seasonal storage*, arXiv:1710.07593v2, 2018.
- [12] Open Data Réseaux Énergies, *Lignes aériennes RTE*, 2024. <https://odre.opendatasoft.com/explore/dataset/lignes-aeriennes-rte-nv/>
- [13] J. Hörsch, F. Hofmann, D. Schlachtberger, and T. Brown, *PyPSA-Eur: An Open Optimisation Model of the European Transmission System*, Energy Strategy Reviews, 22, 207–215, 2018. <https://doi.org/10.1016/j.esr.2018.10.002>
- [14] S. Hagspiel, C. Jägemann, D. Lindenberg, T. Brown, S. Cherevatskiy, and E. Tröster, *Cost-optimal power system extension under flow-based market coupling*, Energy, 66, 654–666, 2014.
- [15] M. Fürsch, S. Hagspiel, C. Jägemann, S. Nagl, D. Lindenberg, and E. Tröster, *The role of grid extensions in a cost-optimal transformation of the European electricity system until 2050*, Applied Energy, 104, 642–652, 2013.
- [16] Danish Energy Agency, *Technology Data for Energy Transport*, July 2025. <https://github.com/PyPSA/technology-data>
- [17] RTE, *Schéma Décennal de Développement du Réseau — Fiche 13 : Choix technologiques*, 2025. <https://assets.rte-france.com/prod/public/2025-03/2025-03-21-sddr-fiche-13-choix-technologiques.pdf>

- [18] RTE, *Open Data API – Production par groupe de production*, 2024. <https://data.rte-france.com>
- [19] IGN / data.gouv.fr, *Communes de France – Codes régionaux et coordonnées géographiques*, 2025. <https://www.data.gouv.fr/fr/datasets/communes-de-france/>
- [20] HiGHS Development Team, *HiGHS – High Performance Software for Linear Optimization*, <https://highs.dev>
- [21] EDF, *Flamanville 3 – Données techniques*, 2024. <https://www.edf.fr/centrale-nucleaire-flamanville3>
- [22] RTE, *Bilan Électrique 2024 – Production nucléaire*, 2025. <https://analysesetdonnees.rte-france.com/bilan-electrique-2024/production>

Appendix

Table 2: Data Sources

Data	Description	Source
National Consumption & Generation	National consumption and generation for 2024, hourly resolution	RTE eCO2mix [6]
Regional Energy Balance	Consumption and generation for 2024, hourly resolution, for the 12 administrative regions of France	RTE eCO2mix régionale [7]
Solar and Wind Capacity Factor	Solar and wind capacity factor by hour for 2024 per region, using modeled time series based on reanalysis weather data (ERA5 / MERRA-2) and the existing installed capacity configuration	Renewables Ninja [9,10]
Transmission Capacity	Inter-regional transmission modeled by aggregating the main 400 kV and 225 kV lines between regions	Open Data Réseaux Énergies [12]
Nuclear Fleet Retirement	List of active nuclear power plants in France with their date of first interconnection and installed capacity	IAEA PRIS [5]
Nuclear Unit Generation	Hourly actual generation per unit for the selected nuclear units in 2024	RTE Open Data API [18]
Regional Centroids	French communes with region codes and coordinates, used to compute inter-regional distances	IGN / data.gouv.fr [19]
Transmission Cost	Annualized capital and operational cost for 400 kV and 225 kV overhead lines	Danish Energy Agency [16]; RTE SDDR [17]
Technology Cost	Annualized capital and operational cost for onshore wind, solar PV, and BESS	RTE <i>Futurs énergétiques 2050</i> [8]